Quantitative Methods for Management

Tutorial Exercise Sheet: Complex Numbers

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Exercise 1 (Complex numbers and basic arithmetic)**.** Simplify each expression to the form $a + bi$ and plot the resulting number in the complex plane. Recall that $i^2 = -1$.

(a) $(-3 + 3i) + (1 - 5i)$

(b) $(2 + i) - (4 - 3i)$

(c) $(1 + 2i)(1 + 3i)$

(d) $\dfrac{3 + i}{1 - i}$

(e) $\dfrac{2 + 4i}{2i}$

(f) $\dfrac{5}{2 + i^5}$

(g) $\dfrac{1 - 3i^3}{2 + i^2}$

(h) $\dfrac{2 + i}{1 - i} + \dfrac{2 - i}{1 + i}$

Solution. For $z = a+bi$, the real part is a and the imaginary part is b. The corresponding point in the complex plane is (a, b). To divide by a complex number, multiply the numerator and denominator by the complex conjugate of the denominator.

(a) Combine real and imaginary parts:

$$(-3 + 3i) + (1 - 5i) = (-3 + 1) + (3 - 5)i = -2 - 2i.$$

(b) Distribute the minus sign and combine terms:

$$(2 + i) - (4 - 3i) = 2 + i - 4 + 3i = -2 + 4i.$$

(c) Expand the product and use $i^2 = -1$:

$$(1 + 2i)(1 + 3i) = 1 + 3i + 2i + 6i^2 = -5 + 5i.$$

(d) Multiply by the conjugate $1 + i$:

$$\frac{3 + i}{1 - i} = \frac{(3 + i)(1 + i)}{(1 - i)(1 + i)} = \frac{2 + 4i}{2} = 1 + 2i.$$

(e) Use $1/i = -i$:

$$\frac{2+4i}{2i} = \frac{1}{i} + 2 = 2 - i.$$

(f) Since $i^4 = 1$, we have $i^5 = i$. Thus,

$$\frac{5}{2+i^5} = \frac{5}{2+i} = \frac{5(2-i)}{(2+i)(2-i)} = \frac{10-5i}{5} = 2 - i.$$

(g) Use $i^2 = -1$ and $i^3 = -i$:

$$\frac{1-3i^3}{2+i^2} = \frac{1+3i}{2-1} = 1 + 3i.$$

(h) Simplify each fraction and add:

$$\begin{aligned}\frac{2+i}{1-i} + \frac{2-i}{1+i} &= \frac{(2+i)(1+i)}{2} + \frac{(2-i)(1-i)}{2} \\ &= \frac{1+3i}{2} + \frac{1-3i}{2} = 1.\end{aligned}$$

Complex numbers in the complex plane

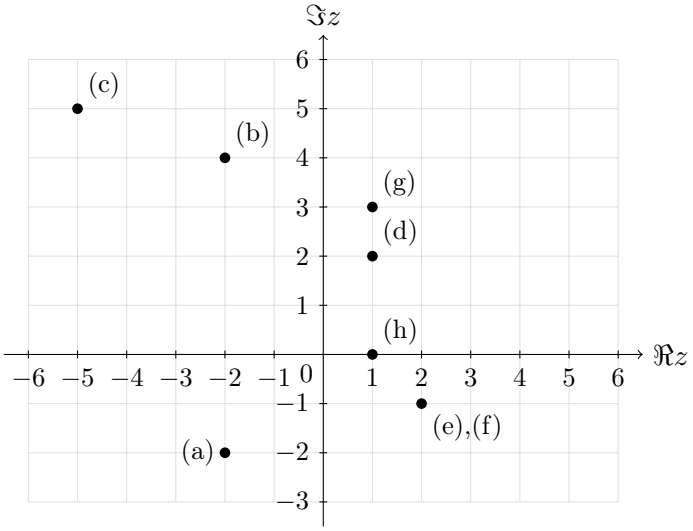


Figure 1: The plot of Exercise 1. Note that Parts (e) and (f) give the same point.

□

Recap: Representations of complex numbers. A nonzero complex number $z \in \mathbb{C}$ can be written in three different forms:

$$\begin{aligned}\text{Cartesian form:} \quad & z = a + bi, \\ \text{Polar form:} \quad & z = r(\cos\theta + i\sin\theta), \\ \text{Exponential form:} \quad & z = re^{i\theta},\end{aligned}$$

where $a, b \in \mathbb{R}$, the radius $r = |z| = \sqrt{a^2+b^2} \geq 0$ is the distance from the origin, and θ is the angle measured counterclockwise from the positive real axis, in radians. The relations

$$a = r\cos\theta, \qquad b = r\sin\theta$$

help determine the angle. Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ connects the polar and exponential forms. Note that angles that differ by an integer multiple of 2π represent the same direction. Thus, it is sufficent to choose $\theta \in [0, 2\pi)$. The number 0 has radius 0 and no defined angle.

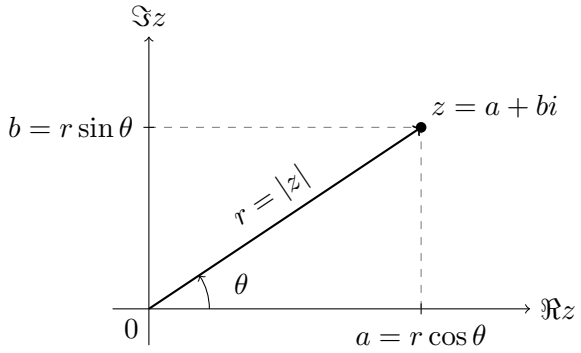


Figure 2: Different representations of a complex number in the complex plain.

Exercise 2 (Polar and exponential representations)**.** Find the radius and angle of each complex number.

(a) $1 + i$

(b) $-1 + i$

(c) $-2i$

Solution. (a) For $1 + i$, the radius is $r = \sqrt{1^2 + 1^2} = \sqrt{2}$. The point $(1, 1)$ is an $\frac{1}{8}$th rotation, $\theta = 2\pi/8 = \pi/4$. Hence,

$$1 + i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = \sqrt{2}e^{i\pi/4}.$$

(b) For $-1 + i$, the radius is $r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$. The point $(-1, 1)$ is an $\frac{3}{8}$th rotation, so $\theta = 6\pi/8 = 3\pi/4$. Hence,

$$-1 + i = \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right) = \sqrt{2}e^{3i\pi/4}.$$

(c) For $-2i$, the radius is $r = \sqrt{0^2 + (-2)^2} = 2$. The point $(0, -2)$ is an $\frac{3}{4}$th rotation, so $\theta = 6\pi/4 = 3\pi/2$. Hence,

$$-2i = 2\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = 2e^{3i\pi/2}.$$

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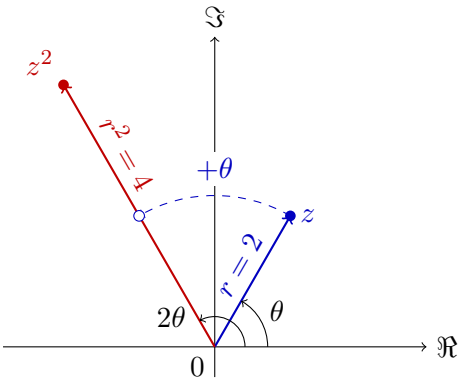
Recap: De Moivre's formula. For $z = r(\cos\theta + i\sin\theta)$ and a positive integer n,

$$\big[r(\cos\theta + i\sin\theta)\big]^n = r^n\big(\cos(n\theta) + i\sin(n\theta)\big).$$

Multiplying a complex number by z rotates its position around the origin by the angle θ and multiplies its radius by r. Starting at 1 on the real axis, multiplying by z repeatedly

gives $z, z^2, z^3, \ldots$ Each step adds θ to the angle and multiplies the radius by r. After n steps, the angle is $n\theta$ and the radius is r^n. Full turns of 2π do not change the final direction. If $r = 1$, each step is a rotation only; if $r > 1$, the radius increases, and if $0 < r < 1$, it decreases.

One multiplication step: from z to z^2



Exercise 3 (De Moivre's formula)**.** Use De Moivre's formula to calculate the following powers. Express each result in the form $a + bi$.

(a) $(1 + i)^4$

(b) $(1 - i)^6$

(c) $(-1 + i)^3$

Solution. Write each number in polar form, raise its radius to the given power, and multiply its angle by that power.

(a) The number $1 + i$ has radius $\sqrt{2}$ and angle $\pi/4$. Therefore,

$$\begin{aligned}(1 + i)^4 &= (\sqrt{2})^4 (\cos \pi + i \sin \pi) \\ &= 4(-1 + 0i) = -4.\end{aligned}$$

(b) The number $1 - i$ has radius $\sqrt{2}$ and angle $7\pi/4$. Multiplying the angle by 6 gives $21\pi/2 = \pi/2 + 5 \cdot 2\pi$. Hence,

$$\begin{aligned}(1 - i)^6 &= (\sqrt{2})^6 \left(\cos \frac{21\pi}{2} + i \sin \frac{21\pi}{2}\right) \\ &= 8 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right) = 8i.\end{aligned}$$

(c) The number $-1 + i$ has radius $\sqrt{2}$ and angle $3\pi/4$. Multiplying the angle by 3 gives $9\pi/4 = \pi/4 + 2\pi$. Hence,

$$\begin{aligned}(-1 + i)^3 &= (\sqrt{2})^3 \left(\cos \frac{9\pi}{4} + i \sin \frac{9\pi}{4}\right) \\ &= 2\sqrt{2} \left(\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}\right) = 2 + 2i.\end{aligned}$$

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Exercise 4 (Polynomial factorization)**.** Factorize the following polynomials over $\mathbb{C}$ and over $\mathbb{R}$. Over $\mathbb{C}$, use linear factors. Over $\mathbb{R}$, use linear factors and quadratic factors that cannot be factored further with real coefficients.

(a) $z^2 - 4z + 5$

(b) $z^3 - 4z$

(c) $z^3 - z^2 + 2$

Solution. (a) Use $z^2 - 4z + 4 = (z-2)^2$ to complete the square, which gives

$$z^2 - 4z + 5 = (z-2)^2 + 1.$$

The roots are $2 + i$ and $2 - i$. Thus, over $\mathbb{C}$,

$$z^2 - 4z + 5 = (z - 2 - i)(z - 2 + i).$$

Over $\mathbb{R}$, the quadratic $z^2 - 4z + 5$ cannot be factored further because it has no real roots.

(b) Factor out z and apply the 3rd Binomial Formula:

$$z^3 - 4z = z(z^2 - 4) = (z - 0)(z - 2)(z + 2).$$

All three roots, 0, 2, and -2, are real, so this is the factorization over both $\mathbb{C}$ and $\mathbb{R}$.

(c) Substituting $z = -1$ gives $(-1)^3 - (-1)^2 + 2 = 0$, so $z + 1$ is a factor. Polynomial division gives

$$z^3 - z^2 + 2 = (z + 1)(z^2 - 2z + 2).$$

Since $z^2 - 2z + 2 = (z-1)^2 + 1$ has no real roots, this is the factorization over $\mathbb{R}$. Its complex roots are $1 + i$ and $1 - i$, so over $\mathbb{C}$,

$$z^3 - z^2 + 2 = (z + 1)(z - 1 - i)(z - 1 + i).$$

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Exercise 5 (Sets in the complex plane)**.** Sketch the following sets in the complex plane. State whether each boundary is included.

(a) $A = \{z \in \mathbb{C} : |z - (1 + i)| \leq 1\}$.

(b) $B = \{\overline{z} : z \in A\}$. Describe how complex conjugation changes the set in part (a).

(c) $C = \{z \in \mathbb{C} : 1 < |z| < 2\}$.

(d) $D = \{z \in \mathbb{C} : \Im(z) > 2\Re(z)\}$.

Solution. Write $z = x + iy$, where $x, y \in \mathbb{R}$.

(a) The inequality $|z - (1 + i)| \leq 1$ describes all points at distance at most 1 from $(1, 1)$. Its boundary is included.

(b) Complex conjugation maps $x+iy$ to $x-iy$, so it reflects A across the real axis. The center becomes $(1, -1)$ and the radius remains 1. Again its boundary is included.

(c) The radius $|z|$ is the distance from the origin. Thus, C is the annulus between the circles of radii 1 and 2 centered at the origin. The inner and outer boundaries are not included.

(d) Since $\Re(z) = x$ and $\Im(z) = y$, the condition becomes

$$y > 2x.$$

Thus, D is the open half-plane above the line $y = 2x$. This line passes through the origin and has slope 2. The boundary is excluded.

Sets in the complex plane: parts (a) and (b)

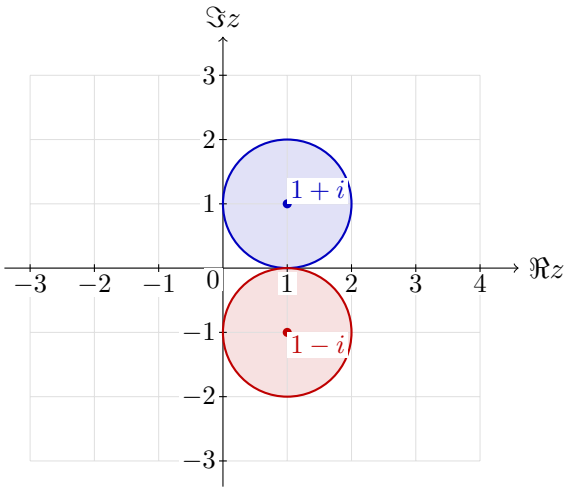


Figure 3: Blue: the disk A in part (a). Red: its conjugate set B in part (b). Complex conjugation reflects the disk across the real axis.

Sets in the complex plane: parts (c) and (d)

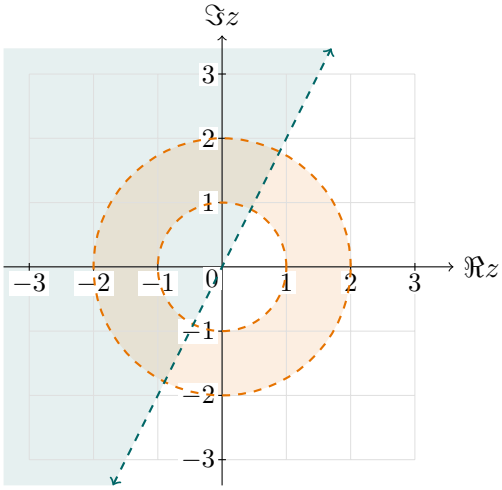


Figure 4: Orange: the annulus C in part (c), with both dashed circular boundaries excluded. Teal: the half-plane D in part (d), where $y > 2x$, with the dashed boundary line $y = 2x$ excluded.

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